

P-Set V

POLI 784 - Spring 2026

2026-03-31

Complete all of the questions below together with your group members. Please submit both (1) a compiled PDF or HTML file with your responses and (2) the **R Markdown**, **Quarto**, etc. file containing both your responses and code as a group. You are encouraged to use **R Studio** projects and virtual environments to complete this p-set, although it is not required. Similarly, you are encouraged to use consistent style and linters when writing code.

Question 1

- a. *(0.5 points)* What is the estimand in a sharp regression discontinuity design (RDD)? How about in a fuzzy RDD?
- b. *(0.5 points)* How do we account for the influence of confounders in a sharp RDD – i.e., what is (are) the identification assumption(s)?
- c. *(0.5 points)* What consequence does fitting global polynomials to estimate the causal effect in an RDD have and what alternative estimation strategy is available to us under continuity?
- d. *(0.5 points)* What consequence does relying on cross-validation to select the optimal bandwidth h in an RDD have and how can we address it?

Question 2

In this exercise, we will replicate the analysis in Goyal and Sells (2023). The authors studied how the victory of female candidates in Brazilian local elections affects party building during their terms. They claim that “women mayors recruit new members at similar rates as men but reduce the gender gap in party membership.” The outcome variable is **Gender.Gap**, defined as “the difference between the recruitment rate of men and the recruitment rate of women in a given municipality.” The forcing variable is the margin of victory. You can find the data needed to solve this exercise in **data1.csv**.

- a. (0.5 points) Show how the outcome varies across the forcing variable using a scatter plot and a regression line. What is the treatment in this study?
- b. (0.5 points) Plot the outcome against the running variable using `rdplot()` with a 4th order polynomial on either side. Why are there fewer data points on this plot than in the one from 2a?
- c. (0.5 points) Estimate the causal effect using `rdrobust()` and print the results to console.
 - i. Interpret the robust estimate of the treatment effect.
 - ii. What are the values of h and b and what are these quantities, respectively?
 - iii. How were h and b selected and why is bias correction necessary?
- d. (0.5 points) Use the selected bandwidth h from 2c to calculate the weight for each observation in the sample using the triangular kernel. Then fit a weighted linear regression model of the outcome on the forcing variable on each side of the cutoff and print the difference (rounded to three decimal places) between their intercepts to console. Is the result consistent with your findings in 2c and why?
- e. (0.5 points) Test the continuity of the density function of the forcing variable across the cutoff using `rddensity()` and `rdplotdensity()`. Show the resulting plot and print the numeric results of the test to console. Do you find any evidence of sorting and why would that be a problem?
- f. (0.5 points) Estimate the local treatment effect for the following placebo tests and interpret their results:
 - i. Use a placebo cutoff by setting c to -0.3 and 0.3 .
 - ii. Use a placebo outcome by applying the estimator to `Gender.Gap.Lagged`.
 - iii. Use a pre-treatment covariate as the outcome by applying the estimator to `UrbanizationRate_2000`.
- g. (0.5 points) Add the covariates `University`, `Married`, `Incumbent`, and `Age` in the RDD estimation. How do the results change and what is the role played by these variables in the estimation?
- h. (0.5 points) Implement the sharp RDD design without covariates using the `msetwo` bandwidth selection algorithm, `hc2` variance estimator, and second-order polynomial local regressions. How do the results change compared to the previous RDD specification without covariates?
- i. (0.5 points) Generate a treatment indicator using `data1$tr <- as.numeric(data1$Margin >= 0)`. Implement a fuzzy RDD design with `tr` as the treatment. Print the results of the estimation to console. What do you notice about the results and how do you explain this?

Question 3

Clingingsmith et al. (2009) studied the impacts of performing the Hajj pilgrimage to Mecca on pilgrims from Pakistan. They utilized the fact that the Pakistani government relies on a lottery to allocate Hajj visas to applicants. The hypothesis they want to test is whether performing the Hajj pilgrimage improves participants' evaluation of people from other countries or groups. One concern the researchers have is non-compliance, as some applicants will head to Mecca even without a Hajj visa from the government, while others may not partake in the pilgrimage despite having a visa. In this replication exercise, we focus on one particular outcome: participants' opinion toward people of a different religion. You can find the data needed to solve this exercise in `data2.csv`.

- a. (0.5 points) Assuming that the instrumental variable satisfies all necessary conditions discussed in lecture:
 - i. Compute the proportion of always-takers, never-takers, and compliers in the sample. Make sure that these proportions sum to 1 and print the results to console.
 - ii. Why are we not computing the proportion of defiers in the sample?
- b. (0.5 point) Estimate the ITT, the LATE using the Wald estimator, and the LATE using the 2SLS estimator (without covariates) and print the results to console.
- c. (0.5 points) Evaluate the strength of the instrumental variable using the first-stage F-test and the Anderson-Rubin test. Print the results to console and interpret your findings.

Question 4

Consider the following schooling experiment conducted in rural areas of a developing country. In the experiment, randomly selected girls are offered the option to attend “community-based schools” located in their village ($Z_i = 1$). There are three actions they might take: staying at home, attending public school, or attending community-based school, where we denote the treatment status as $D_i \in \{0, 1, 2\}$, respectively. The outcome Y_i is their performance in a standardized exam after a year. Without any restrictions, there will be 9 principal strata (shown in the table below) that depend on their choices under $Z_i = 0$ and $Z_i = 1$. For example, in the first row, we have $D_i(Z_i = 0) = 0$ and $D_i(Z_i = 1) = 0$, meaning that this respondent stays at home no matter if she is offered the option or not.

- a. (0.5 points) Let's impose the following assumption of monotonicity: if $D_i(1) \neq D_i(0)$, then $D_i(1) = 2$.
 - i. Explain what this assumption states.

Treatment status	$Z_i = 0$	$Z_i = 1$
$D_i =$	0	0
	0	1
	0	2
	1	0
	1	1
	1	2
	2	0
	2	1
	2	2

- ii. Present the remaining principal strata in a table similar to the one above. (*NOTE: It may be helpful for the following problems to add a column of letters that uniquely identifies the remaining principal strata you have identified post-assumption.*)
- b. (0.5 points) Based on your reduced table from 4a:
- Complete the table below with the principal strata that comprise the respective group formed by crossing D_i and Z_i .
 - Calculate the proportion of respondents in each group (e.g., $P(D_i = 0|Z_i = 0)$, where the proportions *within* levels of Z_i sum to 1) and print the results to console. You can find the data needed to solve this exercise in `data3.csv`.

Treatment status	$Z_i = 0$	$Z_i = 1$
$D_i = 0$	[...]	[...]
$D_i = 1$	[...]	[...]
$D_i = 2$	[...]	[...]

- c. (0.5 points) Calculate the proportion of each principal stratum. Make sure that these proportions sum to 1 and print the results to console.
- d. (0.5 points) Under the assumption of monotonicity, the local average treatment effect (LATE) represents the effect on respondents whose choices are affected by the instrument. Calculate the total proportion of such respondents based on your answer in 4c. Estimate the LATE as the ratio between the ITT estimate (which you can compute using `lm()`) and this proportion and print the results to console.